Clinical Case Scenario No. I



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Clinical Case Scenarios 1

- **A 34-year-old man was brought to the emergency center after being found unresponsive by his family. The family gave a history of the patient, who had consumed a liter of whiskey earlier that evening on a dare by his friends.**

- **On his physical examination, they noticed he had elevated blood pressure and liver function test, Following abdominal Ultrasonography, he was diagnosed with liver cirrhosis, fatty liver, and Hepatosplenomegaly**

Patient history:

- **No family history of jaundice**
- **Not on any medication nor alternative medication**
- **Has not traveled recently**
- **On further questioning, he has been drinking 6 to 8 beers per night, 2 to 3 nights per week, since the age of 25 years**

Q1- Which of the following is the most likely diagnosis in this patient (Diseases considered and why)?

- **Jaundice**
- **Fatty liver**
- **Alcoholic cirrhosis**
- **Chronic inflammation**

Answer: The information gained from the patient's examination indicates Fatty liver (Alcoholic cirrhosis)

Questions:

- What is fatty liver?
- What causes fatty liver?
- How does alcohol lead to fatty liver?
- What are the first signs of liver damage from alcohol?
- What are the side effects of drinking too much alcohol?
- How do you prevent fatty liver? What substances will prevent fatty liver?
- What are Lipotropic factors?

Q2. What is Fatty liver?

- **Fatty liver is the excessive accumulation of fat, primarily TGs, in the liver cells**
- **Fat is mainly stored in AT.**
- **The Liver is not a storage organ for fat, but it contains about 5% fat.**
- **In pathological conditions, this may go up to 25-30% and is known as fatty liver or fatty infiltration of the liver.**
- **When the accumulation of lipids in the liver becomes chronic. Fibrotic changes occur in cells, which may finally lead to cirrhosis and impairment of liver function.**

What causes fatty liver?

Causes of fatty liver

- It occurs in conditions where there is an imbalance between hepatic triglyceride synthesis and the secretion of VLDL.

Conditions that cause fatty liver

1. High-fat diet
2. Starvation, uncontrolled diabetes mellitus, or insulin insufficiency
3. Alcoholism
4. High cholesterol intake
5. Use of certain chemicals
6. Dietary deficiency of essential fatty acids

- ❑ Essential amino acids
- ❑ Vitamin E and selenium
- ❑ Lipotropic factors

How does alcohol lead to fatty liver?

Excessive alcohol intake: Intake of even a moderate amount of ethanol generates too much NADH, resulting in the inhibition of important processes that require NAD^+ (gluconeogenesis and FAs oxidation) and leading to alterations in the following metabolic reaction

Shift to the right of the following conditions

Di(OH) acetone phosphate ⇌ **α glycerol phosphate**

(↑TG synthesis)

and shift to the left of the following

Mala ⇌ **Oxaloacetate**

Produces a relative deficiency of oxaloacetate, and reduced activity of the TCA cycle also increases lipogenesis and cholesterol synthesis from acetyl CoA

What are the first signs of liver damage from alcohol?

Generally, symptoms of alcoholic liver disease include

- **Abdominal pain,**
- **Dry mouth and increased thirst,**
- **Fatigue,**
- **Jaundice,**
- **Loss of appetite, & nausea.**
- **Your skin may look abnormally dark or light.**
- **Your feet or hands may look red.**

What are the side effects of drinking too much alcohol?

Over time, excessive alcohol use can lead to the development of chronic diseases and other serious problems, including:

- **High blood pressure,**
- **Heart disease & stroke,**
- **Liver disease, and**
- **Digestive problems.**
- **Cancer**
- **Erectile dysfunction**
- **Diabetes: There is a high risk of developing diabetes type II**
- **Fetal alcohol syndrome: Consuming alcohol during pregnancy increases the risk of birth defects**
- **Thinning bones**
- **Mental illness**

How do you prevent fatty liver? What substances will prevent fatty liver?

What are Lipotropic factors

Lipotropic factors

The substances that prevent the accumulation of fat in the liver are known as lipotropic factors. So Dietary deficiency of these factors can result in fatty liver

The various lipotropic agents are;

- **Choline**
- **Methionine**
- **Betaine**
- **Vitamin B_{12} and folic acid**